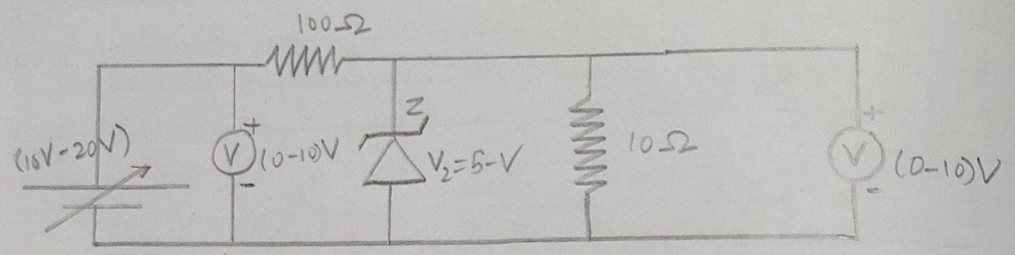
CIRCUIT DIAGRAM

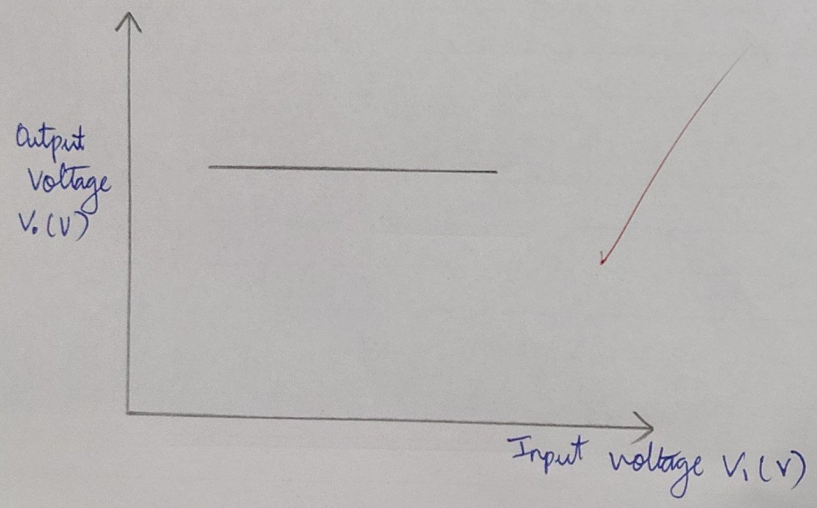
LINE REGULATION



TABULATION:

S.No	INPUT VOLTAGE V_i (V)	OUTPUT VOLTAGE V_o (V)
1	+12.0	+5.14
2	+13.0	+5.16
3	+14.0	+5.17
4	+15.0	+5.17
5	+16.0	+5.18
6	+17.0	+5.19

MODEL GRAPH:



Expt. No. 5

Page No. 40

9/3/26 5. ZENER DIODE AS VOLTAGE REGULATION

AIM:

To stimulate the line and load regulation operation zener diode.

APPARATUS REQUIRED:

Laptop with protens software.

THEORY:

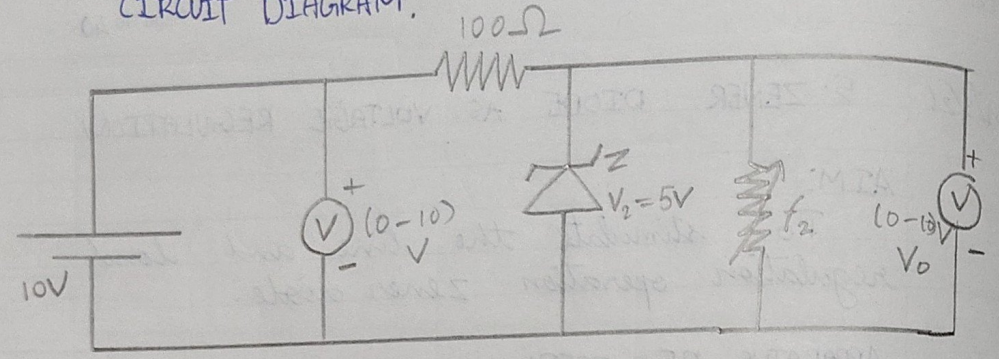
A heavily doped PN Junction diode is called zener diode. Due to heavily dopped nature zener diode works under forward biase and reverse bias conditions.

Zener diode under reverse bias condition is used as a voltage regulation. Even the input voltage changes the output is constant. This is called line regulation.

If the load resistance changes the output voltage is constant this is called load regulation.

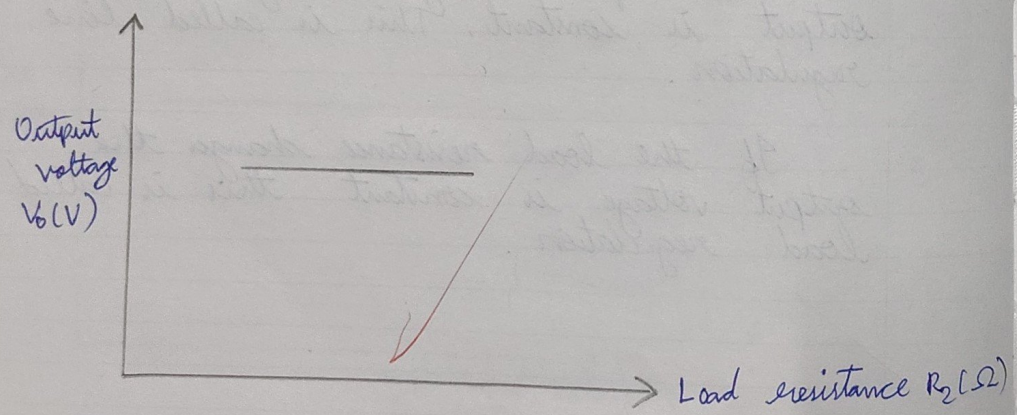
LOAD REGULATION

CIRCUIT DIAGRAM:



TABULATION:

S.No	LOAD RESISTANCE $R_2(\Omega)$	OUTPUT VOLTAGE $V_o(V)$
1.	100Ω	+5.19
2.	200Ω	+5.20
3.	300Ω	+5.20
4.	400Ω	+5.20
5.	500Ω	+5.20
6.	600Ω	+5.21



Expt. No.

PROCEDURE:

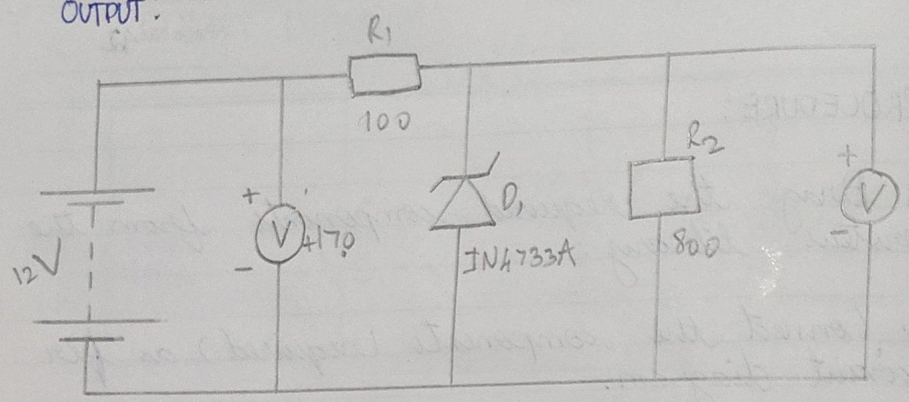
1. Drag the required components from the protenu library.

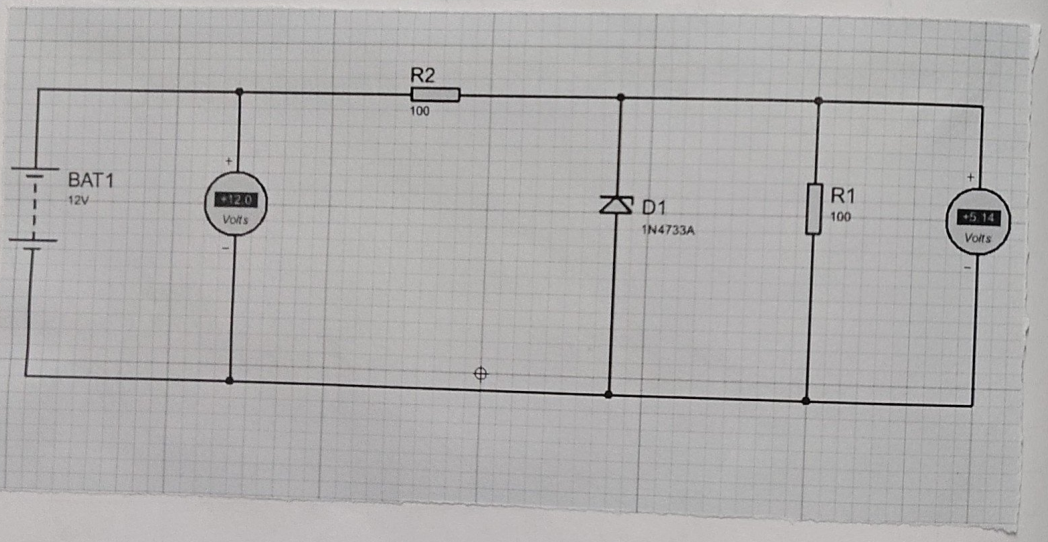
2. Connect the components (required) as per circuit diagram.

3. For line regulation vary the input voltage and not the output voltage in tabulation. Draw the graph between input voltage and output voltage.

4. For load regulation vary the load resistance and note the output voltage in tabulations. Draw the graph between load resistance and output voltage.

OUTPUT :





99

18/03/16

RESULT:

Thus, the stimulation of line load regulation operation of zener diode successfully completed.